WoSM Academy

MAT231-F27 · Calculus II

Vector fields and line integrals

Integrating along curves

Dr. Ali Khudiyev
WoSM Academy

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Last time

- $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$, by a double integral in polar coordinates.
- Improper multiple integrals: exhaustions; absolute convergence; Tonelli and Fubini.
- The normal distribution, $\Gamma(\frac{1}{2}) = \sqrt{\pi}$, and the volume of the ball in $\mathbb{R}^n$.

So far we integrated over intervals, regions and solids. Today: over **curves**. And a new kind of function — one that attaches a vector to each point.

This lecture

01 Vector fields

02 Line integrals of functions

03 Work: line integrals of vector fields

04 The fundamental theorem for line integrals

05 Summary and practice

Reading: Strang & Herman, *Calculus, Volume 3*, Sections 6.1–6.3.

01

Vector fields

Vector fields

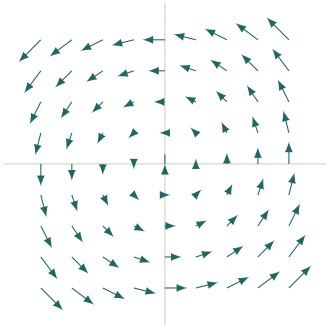
Definition

A **vector field** on a region $D \subseteq \mathbb{R}^n$ is a function $\mathbf{F}\colon D \to \mathbb{R}^n$: it attaches a vector $\mathbf{F}(\mathbf{x})$ to each point $\mathbf{x}$. In the plane, $\mathbf{F} = (P, Q)$; in space, $\mathbf{F} = (P, Q, R)$.

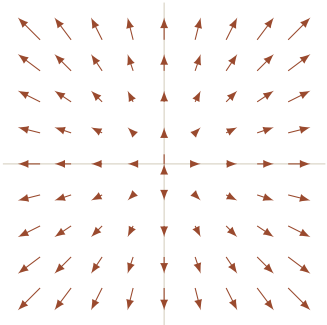
- **Velocity fields**: wind, or the flow of water — the velocity at each point.
- **Force fields**: gravity, electric and magnetic forces.
- **Gradient fields**: ∇f, pointing uphill at each point (Lecture 11). Gradient descent (Lecture 15) follows the field $-\nabla f$ in small steps.

We draw a vector field by drawing some of its arrows, usually scaled down so that they do not overlap.

Two pictures



$\mathbf{F} = (-y, x)$: a **rotation**. $\mathbf{F}$ is perpendicular to the position vector, and $\|\mathbf{F}\| = r$.



$\mathbf{F} = (x, y)$: **radial**, pointing away from the origin. It is the gradient of $f = \frac{1}{2}(x^2 + y^2)$.

Gradient fields

Definition

F is a **gradient field** (or **conservative**) on D if $\mathbf{F} = \nabla f$ for some function f on D, called a **potential** of **F**.

- $(x, y) = \nabla(\frac{1}{2}(x^2 + y^2)), \quad (2xy,\ x^2) = \nabla(x^2 y).$
- Gravity: $\mathbf{F}(\mathbf{x}) = -k\,\dfrac{\mathbf{x}}{\|\mathbf{x}\|^3} = \nabla \dfrac{k}{\|\mathbf{x}\|}$ on $\mathbb{R}^3 \setminus \{\mathbf{0}\}$.

A necessary condition

If $(P, Q) = \nabla f$ with f of class C^2, then $P_y = f_{xy} = f_{yx} = Q_x$ (Schwarz, Lecture 13). So $(-y, x)$ is **not** a gradient field: $P_y = -1$ but $Q_x = 1$.

Flow lines

A **flow line** of $\mathbf{F}$ is a curve $\mathbf{r}(t)$ that follows the arrows: $\mathbf{r}'(t) = \mathbf{F}(\mathbf{r}(t))$.

- $\mathbf{F} = (-y, x)$: $\mathbf{r}(t) = (\cos t, \sin t)$, since $\mathbf{r}'(t) = (-\sin t, \cos t) = \mathbf{F}(\mathbf{r}(t))$. The flow lines are circles around the origin.
- $\mathbf{F} = (x, y)$: $\mathbf{r}(t) = e^t(a, b)$: rays leaving the origin.
- $\mathbf{r}' = \mathbf{F}(\mathbf{r})$ is a system of differential equations; we solve such systems in Lectures 23–25.
- For $\mathbf{F} = -\nabla f$ the flow lines are paths of steepest descent. Gradient descent, $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$, follows them in straight steps of length proportional to η.

02

Line integrals of functions

Curves and arc length

A **smooth curve** C is given by a parametrisation $\mathbf{r}\colon [a,b] \to \mathbb{R}^n$ of class C^1 with $\mathbf{r}'(t) \neq \mathbf{0}$. Its **length** is

$$L = \int_a^b \|\mathbf{r}'(t)\|\, dt$$

— speed integrated over time (in the plane: Calculus I, Unit 24).

> **Example (The helix $\mathbf{r}(t) = (\cos t, \sin t, t)$, $0 \le t \le 2\pi$)**
>
> $\mathbf{r}'(t) = (-\sin t, \cos t, 1)$, $\|\mathbf{r}'(t)\| = \sqrt{2}$, so $L = 2\sqrt{2}\,\pi$.

The length does not depend on the parametrisation: by the chain rule, a change of parameter $t = g(u)$ changes $\|\mathbf{r}'\|\, dt$ to $\|(\mathbf{r}\circ g)'\|\, du$.

Line integrals with respect to arc length

Definition

For f continuous on C:

$$\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,\|\mathbf{r}'(t)\|\,dt.$$

It is the limit of the sums $\sum_i f(\mathbf{r}(t_i^*))\,\Delta s_i$, where Δs_i is the length of the i-th piece of C. Here $ds = \|\mathbf{r}'(t)\|\,dt$ is the element of arc length.

- $f = 1$ gives the length; f a density gives the **mass of a wire**; $\frac{1}{L}\int_C f\,ds$ is the average of f on C.
- It does not depend on the parametrisation, nor on the direction in which C is traversed.

Examples

Example (The upper half of the unit circle, $f = y$)

$\mathbf{r}(t) = (\cos t, \sin t)$, $0 \le t \le \pi$, $\|\mathbf{r}'\| = 1$: $\displaystyle\int_C y\,ds = \int_0^{\pi} \sin t\,dt = 2.$

A semicircular wire of length π has its centroid at height $\bar{y} = \frac{2}{\pi} \approx 0.64$.

Example (The parabola $y = x^2$, $0 \le x \le 1$, $f = x$)

$\mathbf{r}(t) = (t, t^2)$, $\|\mathbf{r}'(t)\| = \sqrt{1 + 4t^2}$:

$$\int_C x\,ds = \int_0^1 t\sqrt{1+4t^2}\,dt = \Big[\frac{(1+4t^2)^{3/2}}{12}\Big]_0^1 = \frac{5\sqrt{5}-1}{12} \approx 0.848.$$

03

Work: line integrals of vector fields

Work done by a force

A constant force $\mathbf{F}$ moving an object along a straight displacement $\mathbf{d}$ does work $W = \mathbf{F} \cdot \mathbf{d}$. Along a curve, cut it into small pieces $\Delta\mathbf{r}_i \approx \mathbf{r}'(t_i)\,\Delta t$ and add: $W \approx \sum_i \mathbf{F}(\mathbf{r}(t_i)) \cdot \mathbf{r}'(t_i)\,\Delta t$.

Definition

The **line integral** of $\mathbf{F}$ along C, oriented by $\mathbf{r}\colon [a,b] \to \mathbb{R}^n$, is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\,dt = \int_C \mathbf{F} \cdot \mathbf{T}\,ds.$$

- $\mathbf{T} = \mathbf{r}'/\|\mathbf{r}'\|$ is the unit tangent: only the component of $\mathbf{F}$ along C counts.
- In coordinates, with $\mathbf{F} = (P, Q)$: $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C P\,dx + Q\,dy$.

Example: around a circle

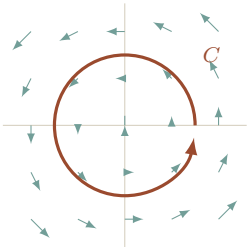
C: the unit circle, counterclockwise,
$\mathbf{r}(t) = (\cos t, \sin t)$, $0 \le t \le 2\pi$.

- $\mathbf{F} = (-y, x)$: $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = (-\sin t, \cos t) \cdot (-\sin t, \cos t) = 1$, so

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi.$$

  The field pushes along the whole way.
- $\mathbf{F} = (x, y)$: $(\cos t, \sin t) \cdot (-\sin t, \cos t) = 0$: no work. The field is perpendicular to the path.

$\oint$ marks a closed curve.



Orientation, and dependence on the path

- Reversing the direction of C changes the sign: $\int_{-C} \mathbf{F} \cdot d\mathbf{r} = -\int_C \mathbf{F} \cdot d\mathbf{r}$ ($\mathbf{T}$ flips). A reparametrisation in the same direction changes nothing.

Does the value depend on the path between two points?

- $\mathbf{F} = (-y, x)$ from $(1,0)$ to $(-1,0)$: along the upper half circle π; along the lower half circle ($\mathbf{r}(t) = (\cos t, -\sin t)$) $-\pi$. **Yes.**
- $\mathbf{F} = (2xy, x^2)$ from $(0,0)$ to $(1,1)$: along $\mathbf{r}(t) = (t,t)$, $\int_0^1 (2t^2 + t^2)\,dt = 1$; along $\mathbf{r}(t) = (t, t^2)$, $\int_0^1 (2t^3 + t^2 \cdot 2t)\,dt = 1$. **The same.**

The second field is a gradient, $\nabla(x^2 y)$, and $1 = x^2 y\big|_{(1,1)} - x^2 y\big|_{(0,0)}$. That is no accident.

04

The fundamental theorem for line integrals

The fundamental theorem for line integrals

Theorem

Let f be C^1 on an open set containing C, a smooth curve from A to B. Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A).$$

Proof.

By the chain rule (Lecture 12), $\frac{d}{dt} f(\mathbf{r}(t)) = \nabla f(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$. So by the Fundamental Theorem of Calculus the integral is $f(\mathbf{r}(b)) - f(\mathbf{r}(a)) = f(B) - f(A)$. □

Compare $\int_a^b F'(x)\,dx = F(b) - F(a)$: the integral of a derivative depends only on the endpoints.

Path independence

- Line integrals of gradient fields are **path independent**, and $\oint_C \nabla f \cdot d\mathbf{r} = 0$ for every closed curve.
- Conversely, if all line integrals of $\mathbf{F}$ on a connected open set are path independent, then $\mathbf{F} = \nabla f$ with $f(\mathbf{x}) = \int_A^{\mathbf{x}} \mathbf{F} \cdot d\mathbf{r}$ (from a fixed point A).
- $(-y, x)$ is not a gradient field: its integral around the circle is $2\pi \neq 0$.

Three equivalent properties of a continuous field F on a connected open set

(a) $\mathbf{F} = \nabla f$ for some f; (b) $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on the endpoints of C; (c) $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed curve C.

Example: gravity

The gravitational force of a mass at the origin, $\mathbf{F} = -k\,\dfrac{\mathbf{x}}{\|\mathbf{x}\|^3}$, is ∇f with $f = \dfrac{k}{\|\mathbf{x}\|}$.

- The work done by gravity moving an object from distance r_1 to distance r_2 is

$$f(B) - f(A) = k\Big(\frac{1}{r_2} - \frac{1}{r_1}\Big),$$

  whatever the path. Moving outwards ($r_2 > r_1$), it is negative: we must work against gravity.
- In physics, $U = -f$ is the **potential energy**, and "conservative" refers to the conservation of kinetic plus potential energy in such fields.

Finding a potential

Example ($\mathbf{F} = (2xy + 1,\ x^2 + 3y^2)$)

1. Test: $P_y = 2x = Q_x$. (Necessary.)
2. Integrate P in x: $f = x^2y + x + g(y)$, with g unknown.
3. Match Q: $f_y = x^2 + g'(y) = x^2 + 3y^2$, so $g(y) = y^3$.

So $f = x^2y + x + y^3$. Along **any** curve from $(0,0)$ to $(1,2)$:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(1,2) - f(0,0) = 2 + 1 + 8 = 11.$$

A warning: holes

Example ($\mathbf{F} = \dfrac{(-y,\ x)}{x^2+y^2}$ on $\mathbb{R}^2 \setminus \{\mathbf{0}\}$)

A computation gives $P_y = Q_x = \dfrac{y^2 - x^2}{(x^2+y^2)^2}$. But around the unit circle, $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = 1$, so $\oint_C \mathbf{F} \cdot d\mathbf{r} = 2\pi \neq 0$: **not** a gradient field.

- Near each point, $\mathbf{F} = \nabla\theta$, the gradient of the polar angle. But θ cannot be defined continuously all the way around the origin: it gains 2π.
- So $P_y = Q_x$ is necessary, but not sufficient. It is sufficient on regions without holes (such as $\mathbb{R}^2$ or a disc); the proof uses Green's theorem.

Green's, Stokes' and the divergence theorem relate integrals over curves, surfaces and solids (Marsden & Tromba, Chapter 8).

05

Summary and practice

Summary

- Vector fields $\mathbf{F}\colon D \to \mathbb{R}^n$; gradient fields ∇f; flow lines $\mathbf{r}' = \mathbf{F}(\mathbf{r})$.
- Arc length $\int_a^b \|\mathbf{r}'\|\,dt$; $\int_C f\,ds = \int_a^b f(\mathbf{r}(t))\,\|\mathbf{r}'(t)\|\,dt$ (mass, averages), independent of direction.
- Work: $\int_C \mathbf{F}\cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t))\cdot\mathbf{r}'(t)\,dt$; it changes sign with the direction, and in general depends on the path.
- **Fundamental theorem**: $\int_C \nabla f\cdot d\mathbf{r} = f(B) - f(A)$. Gradient fields are exactly the fields with path-independent integrals.
- Test in the plane: $P_y = Q_x$ (necessary; sufficient on regions without holes).

Practice

1. Is (y, x) a gradient field? Is $(y, -x)$? Find a potential where there is one.
2. Compute $\int_C (x + y)\, ds$ for the segment from $(0, 0)$ to $(3, 4)$.
3. Find the mass of the helix $\mathbf{r}(t) = (\cos t, \sin t, t)$, $0 \le t \le 2\pi$, with density z.
4. For $\mathbf{F} = (x^2, xy)$, compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 1)$ along $y = x$ and along $y = x^2$. Is $\mathbf{F}$ a gradient field?
5. Find a potential for $\mathbf{F} = (ye^{xy},\ xe^{xy} + 2y)$, and compute $\int_C \mathbf{F} \cdot d\mathbf{r}$ from $(0, 0)$ to $(1, 1)$.
6. Compute the work done by $\mathbf{F} = (-y, x)$ around the square with vertices $(\pm 1, \pm 1)$, counterclockwise.
7. Show that $\oint \frac{-y\, dx + x\, dy}{x^2 + y^2} = 2\pi$ around the circle of radius R, for every $R > 0$.

Answers

1. $(y, x) = \nabla(xy)$. $(y, -x)$: $P_y = 1 \neq -1 = Q_x$, not a gradient field.
2. $\mathbf{r}(t) = (3t, 4t)$, $\|\mathbf{r}'\| = 5$: $\int_0^1 7t \cdot 5\, dt = \frac{35}{2}$.
3. $\int_0^{2\pi} t\sqrt{2}\, dt = 2\sqrt{2}\,\pi^2$.
4. Along $y = x$: $\int_0^1 2t^2\, dt = \frac{2}{3}$; along $y = x^2$: $\int_0^1 (t^2 + 2t^4)\, dt = \frac{11}{15}$. Different values, so not a gradient field ($P_y = 0 \neq y = Q_x$).
5. $f = e^{xy} + y^2$: $f(1,1) - f(0,0) = e + 1 - 1 = e$.
6. Each side contributes 2: total 8. (Green's theorem: $2 \times$ area.)
7. $\mathbf{r}(t) = (R\cos t, R\sin t)$: the integrand is $\frac{R^2 \sin^2 t + R^2 \cos^2 t}{R^2} = 1$, so 2π.

Next time

Lecture 23 — First-order differential equations (Unit 23)

- What a differential equation is; initial value problems; direction fields
- Separable equations: growth, decay, the logistic equation
- First-order linear equations and integrating factors

Reading: Strang & Herman, *Calculus, Volume 2*, Sections 4.1 and 4.3–4.5.

Questions: the module channel on Discord, or office hours, Monday–Friday, 10:00–17:00 (Baku time, UTC+4), ali.khudiyev@wosm.academy.